

Daehurn Kang

C# and Unity developer · .NET, Python, SQL

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PROFILE

I build games and desktop tools in C#, and I'm looking for a junior software developer role. Since late 2024 I've been making Memora, a mobile rhythm game in Unity, together with the charting editor I use to build its levels, and I've written and released two Windows utilities in .NET 8. I also use Python and SQL for data work, including a pipeline from a rate-limited REST API into MySQL. Two years of consultancy work at Infotection and Xander Talent have made me comfortable working from requirements alongside engineers and non-technical stakeholders. I have an MSci in Chemistry from Imperial College London.

PROJECTS

Memora, a mobile rhythm game and charting editor (Unity 6, C#) Late 2024 – present

- Designed the note model so that the data that decides a note's score (its beat, target and action) is stored separately from its keyframed animation, which means visual effects and camera movement can never change how a note is judged.
- Derived all gameplay timing from the audio DSP clock with scheduled playback, sampling every pose from the current beat instead of integrating per frame, and wrote a four-tap latency calibration that rejects inconsistent taps.
- Built notes that aim at other notes or at moving points, with every target computed as a pure function of chart and beat so that the game and the editor produce identical results when seeking in either direction.
- Built a charting editor in Unity that previews the gamefield at any beat, using evaluators shared with the game, an interval tree over object lifetimes and latest-request-wins seeking.
- Profiled an editor slowdown and cut the average repaint on the largest chart from about 120 ms to 15 ms, confirming no change in output with 18,125 automated comparisons and pixel-identical screenshots.

DesktopIdle, a Windows tray utility (C#, .NET 8, Win32) 2026

- Clears the desktop after a period of inactivity and restores it when the user returns, using Win32 idle detection, window and taskbar control and the Windows media-session API, with a grace period so the app's own cursor movement isn't mistaken for the user coming back.

Touch FX, a click-effect overlay (C#, WPF) 2026

- A transparent, click-through overlay across every monitor that detects clicks with a low-level mouse hook and only subscribes to rendering while an effect is on screen.

League of Legends data study (Python, SQL, Riot Games API) 2024

- Pulled match data from a rate-limited API with throttling and resumable caching, reshaped it to match a public 2020 dataset, loaded both into MySQL with SQLAlchemy and compared them.

EXPERIENCE

Business Analyst (Change), Infotection Oct 2024 – present

- Write technical and business requirements, user stories and acceptance criteria in Azure DevOps, turning broad client goals into specific, testable behaviour for the engineering team to build against.
- Interview business owners and engineers during discovery, and make sure both groups come away with the same understanding of what is being built.
- Map how processes work now and how they should work afterwards, including the steps where data is brought in and validated.
- Run research sessions with the people who manage records day to day, and support rollouts with training and follow-up once a change has gone live.

Technical Consultant, Xander Talent

Dec 2023 – 2024

- Redesigned an ETL process so that it no longer collected duplicate form fields or extracted data nobody used, and presented the results to the client.
- Looked into why a client was making so little use of their Salesforce CRM, comparing how they worked at the time with how they could work, and recommended what would need to change.
- Put together short reports from the client's data that people without a technical background could act on.
- Worked in Scrum sprints with regular reviews alongside the client.

US/UK Tax Trainee, Jaffe & Co

Jul – Nov 2023

- Prepared and checked US and UK tax returns for individuals and companies, which meant working to two sets of rules at once.
- Reviewed client documents to make sure they were complete and valid before anything was filed.

Computational Research Assistant, University of Cambridge · In2Research

Jul – Sep 2022

- Wrote a Python package that generates random polyethylene structures and relaxes them with a force field, following the AIRSS structure-search method.
- Presented the results as a poster at the In2Research summer conference.

EDUCATION

MSci Chemistry (2:1), Imperial College London

2019 – 2023

My dissertation looked at how reproducible it is to build Gaussian16 inputs for large biological assemblies, and I developed a workflow to make the process less error-prone.

Technical specialism, Xander Academy

2023 – 2024

A technical training programme covering Python (Flask, SQLAlchemy and Pandas), SQL with MySQL, data modelling, ETL and Power BI.

A-levels, LaSWAP Sixth Form

2017 to 2019

Chemistry (A*), Computer Science (A), Mathematics (A), Physics (A).

SKILLS

Development	C# (Unity 6, .NET 8, WinForms, WPF), Unity editor tooling, Win32 interop, JSON data formats, Git and GitHub, profiling
Data	Python (Pandas, NumPy, Requests), SQL (MySQL, SQLAlchemy), REST APIs, Tableau, Power BI
Delivery	Requirements and acceptance criteria, Azure DevOps, Scrum
Languages	English, Korean (fluent)